

Software Engineering for CS

Week 1

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Switch off mobile phones
during lectures,
or
put them into silent mode

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- What is Engineering?
- What is a Software?
- What do you understand by Software Engineering?

Software

Computer software, or just **software**, is a collection of [computer programs](#) and related [data](#) that provide the instructions for telling a [computer](#) what to do and how to do it

Software is a set of **programs, procedures, algorithms** and its **documentation**

Reference :

http://en.wikipedia.org/wiki/Computer_software

Software Engineering (1)

Software Engineering is a discipline concerned with the development of software systems by applying engineering principles with the goal of developing cost-effective quality systems. IEEE defines software engineering [IEEE-93] as

- The application of systematic, disciplined, quantifiable approach to development, operation, and maintenance of software; that is application of engineering to software.
- The study of approaches mentioned above.

The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software. [IEEE]

Software Engineering (2)

“all aspects of software production”

Software engineering is not just concerned with the technical processes of software development but also with activities such as software project management and with the development of tools, methods and theories to support software production.[Sommerville]

Software Engineering is not just about producing software, but about producing software in a COST-EFFECTIVE manner

The Discipline of Software Engineering

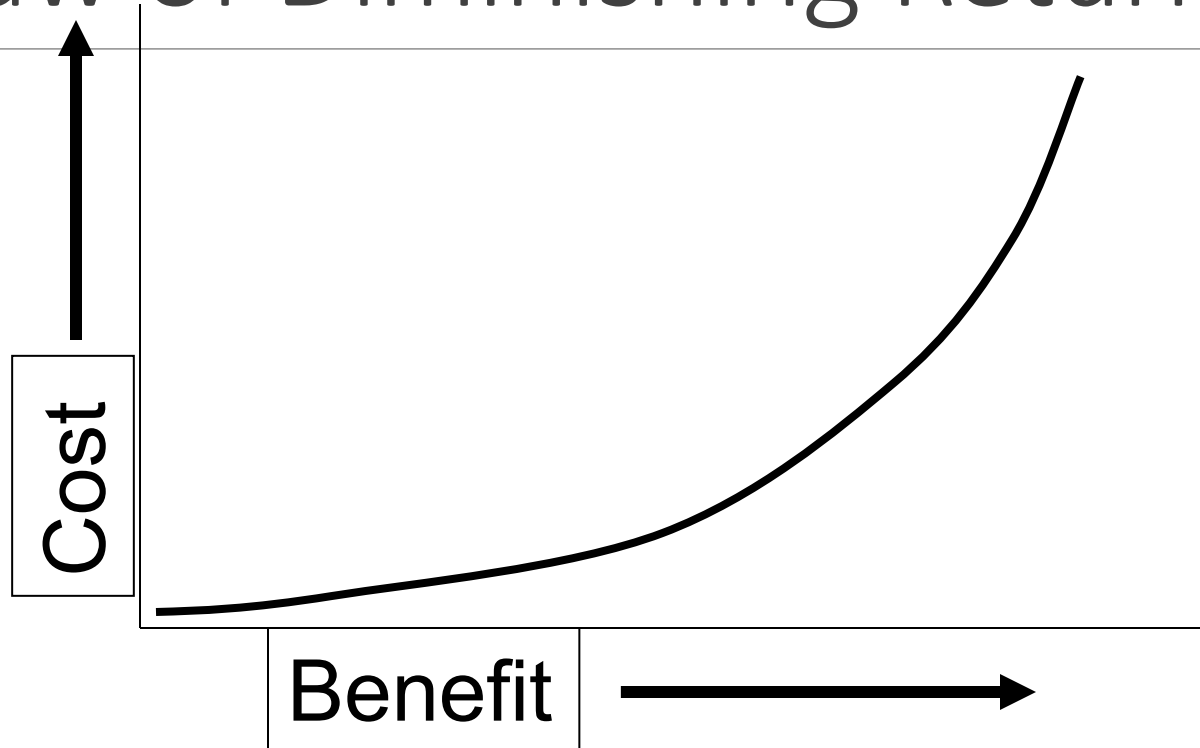
It is a discipline of creating high-quality software in a systematic, controlled and efficient manner, and maintaining it affordably. It involves the application of engineering concepts, techniques, and methods to the development of software systems.

Software Engineering is the discipline of developing and maintaining software systems that behave reliably and efficiently, and are affordable to develop and maintain. However, more recently it has evolved in response to the increased importance of software in safety-critical applications and to the growing impact of large and expensive software systems in a wide range of situations.

Software Engineer

A software engineer is challenged to produce high-quality software with finite amount of resources and to a predicted schedule and budget

Law of Diminishing Returns



Making the engineering decision!

The Balancing Act!

Potentially conflicting requirements

- Cost vs. Efficiency
- Cost vs. Reliability
- Efficiency vs. User-interface

Challenge is to balance these requirements

Well-Engineered Software

Provides the required functionality

- Reliable
- User-friendly
- Cost-effective
- Efficient
- Maintainable

Breakdown of Effort Cost - Software Cost Estimation [Capers Jones]

No.	Activity	Effort		
1	Requirements	3.84%	13	Configuration Mgt. 0.41%
2	Prototyping	4.50%	14	Integration 2.71%
3	Architecture	2.25%	15	User Documentation 9.67%
4	Project Plans	1.33%	16	Unit Testing 4.50%
5	Initial Design	3.84%	17	Function Testing 4.50%
6	Detailed Design	4.50%	18	Integration Testing 3.84%
7	Design Reviews	3.02%	19	System Testing 3.38%
8	Coding	13.50%	20	Beta Testing 3.02%
9	Reuse acquisition	1.13%	21	Acceptance Testing 1.94%
10	Package Purchase	1.69%	22	Independent Testing 3.38%
11	Code Inspection	4.50%	23	Quality Assurance 4.50%
12	Independent V&V	5.42%	24	Installation and Training 1.94%
			25	Project Management 6.75%

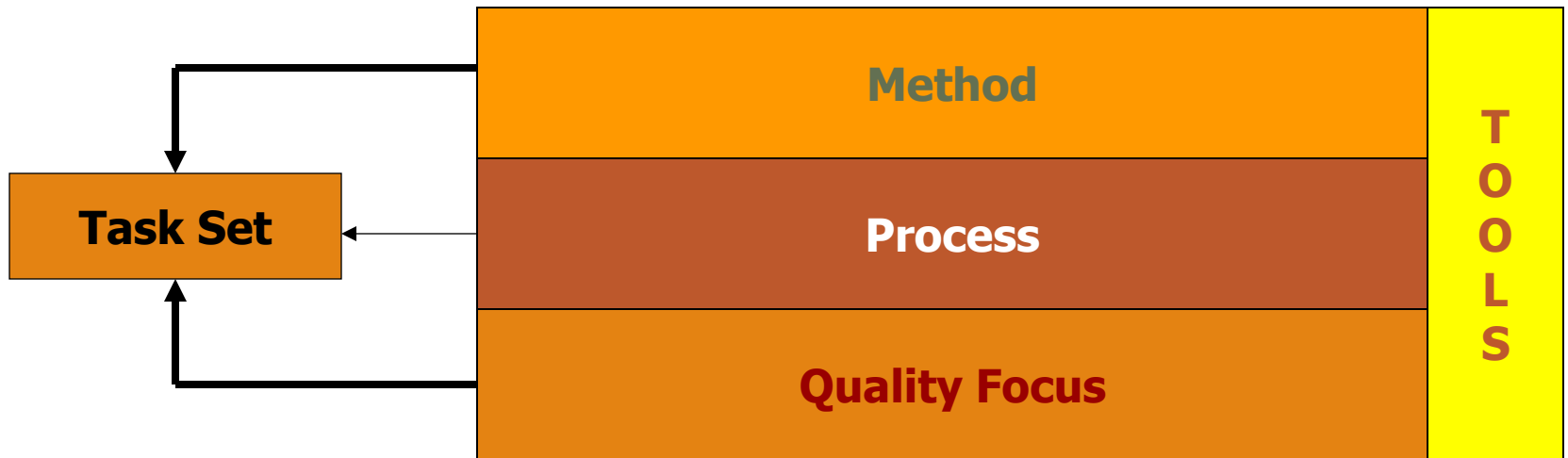
A Software Engineering Framework

- Any Engineering approach must be founded on organizational commitment to quality.
- The quality focus demands that processes be defined for rational and timely development of software. The processes a framework for a set of key process areas (KPAs) for effectively manage and deliver quality software in a cost effective manner.
- The processes define the tasks to be performed and the order in which they are to be performed along with the milestones and deliverables at these milestones.
- Methods provide the technical “how-to’s” to carry-out these tasks.

A Software Engineering Framework

- The quality control defines the SQA points within those tasks and the quality parameters.
- Tools provide automated or semi-automated support for software processes, methods, and quality control.
- The difference between a tool and a method:
 - Standard deviation is a method to estimate and measure dispersion of data
 - Excel is a tool that can be used to calculate standard deviation

A Software Engineering Framework



Thanks

